

AERPAW Tools Collection

Resources

<https://jwcn-urasipjournals.springeropen.com/articles/10.1186/s13638-024-02386-0>

Digital twin based DDPG reinforcement learning for sum-rate maximization of AI-UAV communications - EURASIP Journal on Wireless Commun
Construction of wireless infrastructure using unmanned aerial vehicle (UAV) can effectively expand the coverage and support high-density traffic of next-generation systems. Designing wireless systems including UAVs as aerial base stations (ABSs) is a challenging task, due to the mobility of ABSs causing time-varying nature of surroundings and relative propagation paths to user equipment (UE) devices. Therefore, it is essential to have an accurate estimate of the channel for varying positions.  <https://jwcn-urasipjournals.springeropen.com/articles/10.1186/s13638-024-02386-0>

1. Blender

a. Used for importing topology data

- i. From
- ii. Google Maps
- iii. OpenTopography

1. <https://opentopography.org/>

Home | OpenTopography
 <https://opentopography.org/>

iv. OpenStreetMap

1. <https://www.openstreetmap.org/#map=4/38.01/-95.84>

OpenStreetMap
OpenStreetMap is a map of the world, created by people like you and free to use under an open license.
 <https://www.openstreetmap.org/#map=4/38.01/-95.84>



2.

v. Other LiDAR map?

1. <https://sdd.nc.gov/>
2. <https://data.usgs.gov/datacatalog/data/USGS:63501ed5d34e47431c15c189>

b. BlenderGIS

c. Mistsuda Blender

d. How to import data:

- i. https://www.youtube.com/watch?v=MjZZ1P2hUWk&ab_channel=CGGeek
- ii. https://www.youtube.com/watch?v=6so8Ov0UE1k&ab_channel=NikKottmann
- iii. https://www.youtube.com/watch?v=I4DB9AkIJDo&ab_channel=jbdtube
- iv. https://www.youtube.com/watch?v=-ubDqXZtdJE&ab_channel=MarkAlloway

2. SUMO

a. Used for simulate traffic

- b. Control UE behaviors
- c. <https://sumo.dlr.de/docs/index.html>

3. Sionna

- a. Nvidia
- b. Based on TensorFlow and Python
- c. Simulate wireless channel by ray-tracing
- d. <https://github.com/NVlabs/sionna>
- e. <https://developer.nvidia.com/sionna>

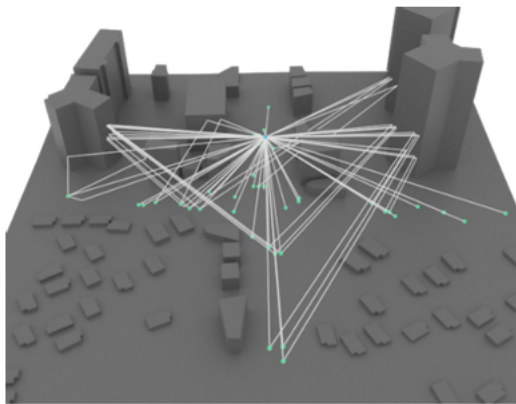
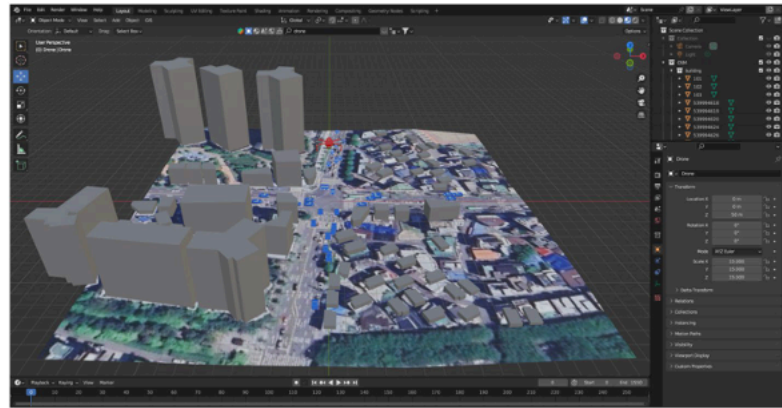


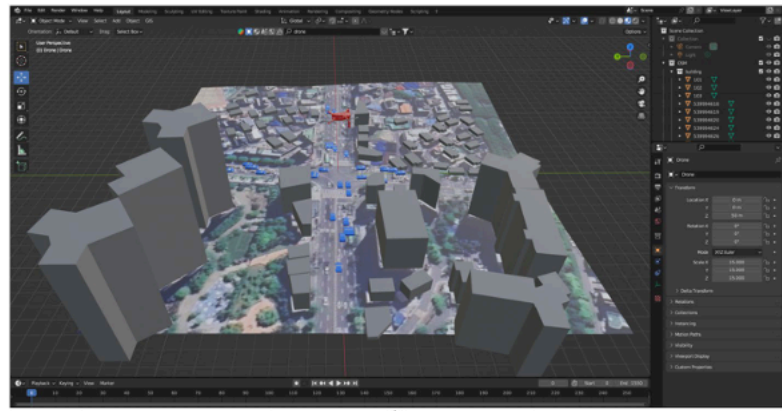
Fig. 3 Illustration of ray-tracing for channel coefficient generation inside the digital twin

4. RL

- a. Single agent, seems simple

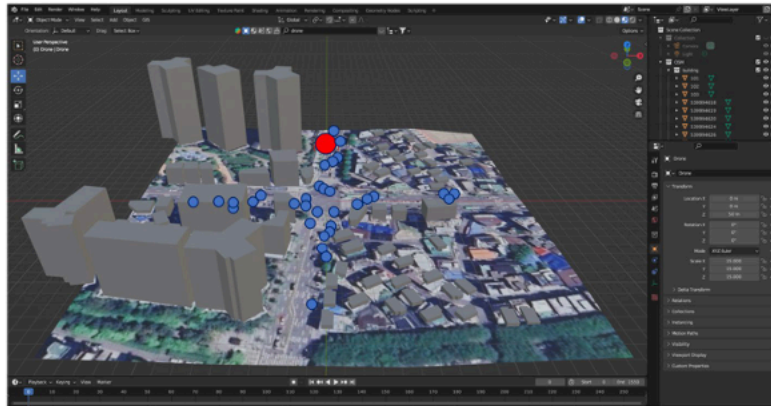


(a)

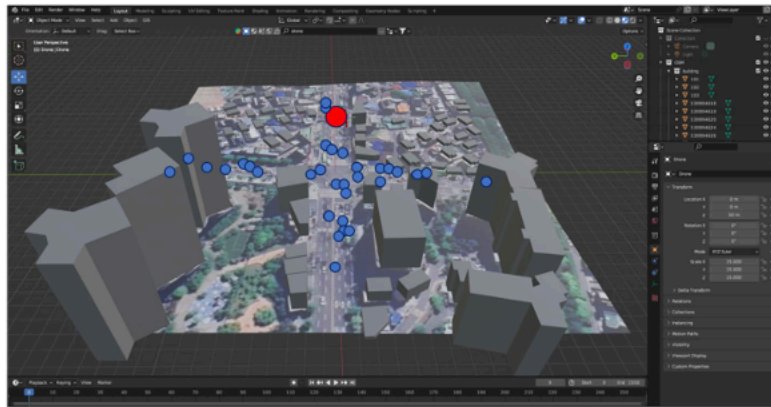


(b)

Fig. 2 The digital twin construction for the intersection near Sogang University Campus: **a** South view, **b** west view, **c** south view with *Tx/Rx* locations indicated by circles, **d** west view with *Tx/Rx* locations indicated by circles



(c)



(d)

Fig. 2 continued

1. Unity

a. Struct and control a in-door DT